

Mr. GUO CHENG

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Education

Purdue University, Ph.D. in Computer Science	08/2016 - 12/2022
• <i>Research Interest:</i> Computer Vision, Machine Learning, Image/Video Processing, Autonomous Driving	
Stony Brook University, M.S. in Statistics	08/2013 - 05/2015
Wuhan Textile University, B.S. in Information and Computer Science	09/2009 - 06/2013

Work Experiences

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- **Tenure-Track Assistant Professor** of Computer Science at **Texas A&M University – Corpus Christi**, Corpus Christi, TX 08/2026 -
 - Research Interests: Computer Vision, Autonomous Driving, Deep Learning
 - **Software Engineer** at **Bloomberg**, New York, NY 10/2022 - 04/2025
 - Trade processing and management of regulatory reports. My work enhanced latency-control for post-trade processing via routing functions with new attributes. (*Backend, C++*)
 - Developed dynamic data models for trading regulatory system. Automated the analytics lifecycle from raw data to actionable intelligence in trading. (*Machine Learning, Python*)
 - Developed and maintained Lare Language Models (LLMs) for Bloomberg law document understanding. Integrated closed-source APIs and fine-tuned open-source models (*PyTorch, AWS*).
 - **Data Scientist Intern** at **ClearObject**, Indianapolis, IN 05/2021 - 08/2021
 - Object detecting/tracking for damage of wooden board in surveillance. (*Computer Vision*)
 - Designed encoder-decoder deep learning models and built the code base for project of ClearVision for Pipes: bounding-box based object detection and data visualization. (*Tensorflow v1/v2, Python*)

Selected Publications

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- **G. Cheng**. Semantic Misalignment in Vision-Language Models under Perceptual Degradation. *arXiv*: 2601.08355.
 - **G. Cheng**. Sequential Semantic Segmentation of Streaming Scenes for Autonomous Driving. (Doctoral Dissertation at Purdue University, Dec. 2022)
 - **G. Cheng**, J. Zheng. Shift-Memory Network for Temporal Scene Segmentation. *arXiv*: 2202.08399.
 - **G. Cheng**, J. Zheng. Sequential Semantic Segmentation of Road Profiles for Path and Speed Planning. IEEE Transactions on Intelligent Transportation System (T-ITS)
 - **G. Cheng**, J. Zheng. SE3: Sequential Semantic Seg of Large Images with Min Memory. (ICPR 2022)
 - **G. Cheng**, J. Zheng. Semantic Segmentation for Pedestrian Detection from Motion in Temporal Domain. International Conference on Pattern Recognition (ICPR 2020)
 - **G. Cheng**, J. Zheng, M. Kilicarslan. Semantic Segmentation of Road Profiles for Efficient Sensing in Autonomous Driving. IEEE Intelligent Vehicles Symposium (IV 2019)
 - **G. Cheng**, J. Zheng. Sparse Coding of Weather & Illuminations for ADAS and Auto Driving. (IV 2018)
 - **G. Cheng**, Z. Wang, J. Zheng. Modeling Weather and Illuminations in Driving Views Based on Big-Video Mining. IEEE Transactions on Intelligent Vehicles (T-IV)
 - **G. Cheng**, Z. Wang, J. Zheng. Big-Video Mining of road appearances in full spectrums of weather and illuminations. IEEE International Conference on Intelligent Transportation Systems (ITSC 2017)
 - Z. Wang, **G. Cheng**, J. Zheng. Planning autonomous driving with compact road profiles. (ITSC 2021)
 - Z. Wang, **G. Cheng**. Road edge detection in all Weather & Illumination via driving video mining.(T-IV)
 - T. Chen, **G. Cheng**, S. Shi. Straight Edge Extraction from LiDAR Images. (ICCECT 2012)

Skills & Languages

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- Python (TensorFlow/PyTorch, HuggingFace, LangChain, RAG), C++, JavaScript/TypeScript